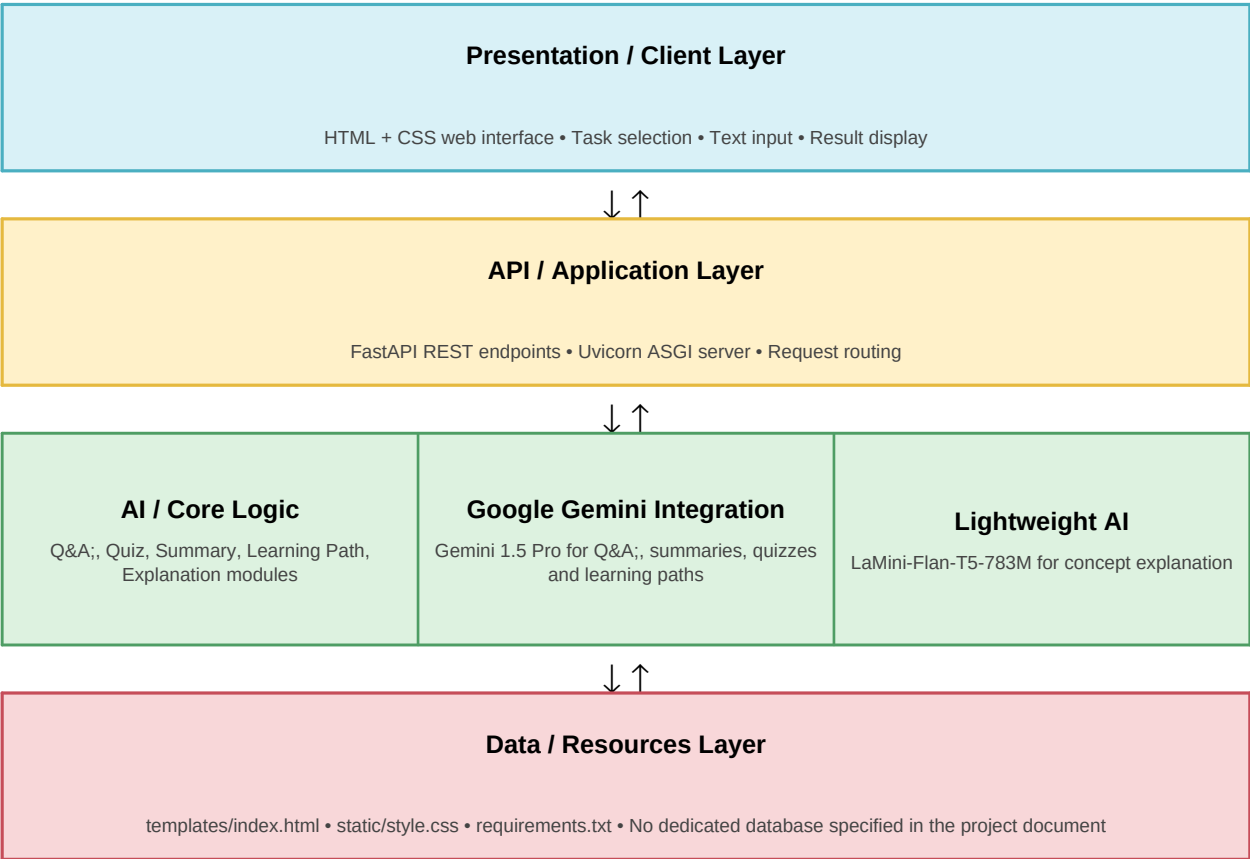


Solution Architecture

Date	05 October 2026
Team ID	SWJID-2026-8809
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram

A high-level representation of EduGenie's architecture based on the supplied project document. The system uses a simple HTML/CSS client, a FastAPI backend, Google Gemini for major generative-AI tasks, and LaMini-Flan-T5 for lightweight concept explanations.



Main API Endpoints

/qa	Question answering using Gemini
/explain	Concept explanation using the lightweight local model
/quiz	Generate three MCQs with four options each
/summarize	Summarize long educational passages
/learn/recommendations	Generate a structured beginner-to-advanced learning path

Solution Architecture — Component & Team Details

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Learner-facing web interface with task selection, text input, submit action and result display.	HTML, CSS
API / Backend Layer	Receives client requests, routes them to the appropriate module, and exposes RESTful endpoints.	Python, FastAPI, Uvicorn
AI / Core Logic	Handles explanation, Q&A;, quiz generation, summarization and personalized learning recommendations.	Gemini 1.5 Pro, LaMini-Flan-T5-783M
Resources / Storage	Contains frontend templates, CSS/static assets and Python dependency definitions. No persistent database is specified in the supplied document.	Jinja2, HTML/CSS, requirements.txt

Team Members & Assigned Work

Member	Assigned Work
1. Divya.K	Frontend Development — HTML/CSS interface, task dropdown, input form, result display and responsive UI.
2. Veronica	Backend Development — FastAPI application, REST endpoints and connection between the frontend and AI modules.
3. Sangeetha.T	AI Module Development — Gemini integration for Q&A;, quiz, summarization and learning-path generation; concept explanation logic.
4. Pooja.	Testing & Integration — functional testing of Q&A;, explanation, quiz, summary and recommendation features; integration checks.
5. Monika.	Documentation & Deployment — project documentation, setup instructions, requirements, local execution and final presentation support.

Project Flow

Learner → HTML/CSS Interface → FastAPI REST API → Appropriate AI Module → Gemini / LaMini-Flan-T5 → Result returned to the learner. The supplied project document describes local execution with Uvicorn and functional testing for all five learning tasks.

Source basis: Prepared from the supplied EduGenie project document and the provided Solution Architecture template. The architecture and work allocation are aligned to the modules, technologies and workflow described in the source.